BBD GLUING OF UNBOUNDED DERIVED OBJECTS OVER AN ARBITRARY OPEN COVER WITH VANISHING NEGATIVE EXT

THE STACKS PROJECT AUTHORS

Editorial note. The original theorem credits BBD Theorem3.2.4; this source proves the stated special case without boundedness. All source-local gluing lemmas used for the count are included as solution support, not separately counted.

Editorial note. At11432 the invoked property is uniqueness of an already existing solution on the limit cover, established by lemma-uniqueness11187-11221; the printed finite-case reference is unnecessary for this uniqueness assertion.

Editorial note. At11430-11435 the unique isomorphism is printed K_alpha→C while the chosen representative has direction C→I_alpha; the latter represents its inverse. The original directions are retained and explicitly identified.

Editorial note. At11546 the final rho indices are reversed in print; the compatibility is the one explicitly imposed in11353-11356 and stated in the preceding composition. Source formulas are retained.

Editorial scope. One original target; complete source-local proof chain included, standard derived-sheaf primitives explicitly delimited.

Allowed standard prerequisites (editorial).

- Standard derived sheaf operations. K-injective resolutions, extension by zero for open embeddings, compatibility of derived pushforward with restriction, and exactness of filtered colimits on stalks.
- Derived Hom and Mayer-Vietoris. Derived internal Hom computes local Ext and H^0 computes derived-category Hom; two-open Mayer-Vietoris gives the exact Hom sequence.
- Elementary topology of a subcollection of opens. A covering family stable under intersections up to unions is the inverse image of a basis under a continuous map; take the topology generated by this family on the underlying set.

Source reference. Original lemma 08BW: focused-seeds/ACL/sources/stacks-snapshot-20261011/cohomology.tex, source lines 8166–8184. Standard two-open exact Hom sequence; the gluing object and its compatibility are constructed locally. Displayed pointer is the original stable Stacks tag.

Source reference. Original lemma 08DK: focused-seeds/ACL/sources/stacks-snapshot-20261011/cohomology.tex, source lines 10580–10589. Standard identification of local derived Hom with H^0 of internal Hom. Displayed pointer is the original stable Stacks tag.

Source reference. Original lemma 0D5X: focused-seeds/ACL/sources/stacks-snapshot-20261011/cohomology.tex, source lines 7941–7949. Standard cohomology-sheaf description for derived direct images. Displayed pointer is the original stable Stacks tag.

Date: stacks-project:cohomology.tex:local-snapshot-20261011; GNU FDL 1.2 or later.

Source reference. Original lemma 0D5Q: focused-seeds/ACL/sources/stacks-snapshot-20261011/topology.tex, source lines 376–390. Elementary topology change providing a basis, explicitly permitted and not a derived descent theorem. Displayed pointer is the original stable Stacks tag.

Situation 0.1. Let $(X, \mathcal{O}_X)$ be a ringed space. We are given

(1) a collection of opens $\mathcal{B}$ of X,
(2) for $U \in \mathcal{B}$ an object K_U in $D(\mathcal{O}_U)$,
(3) for $V \subset U$ with $V, U \in \mathcal{B}$ an isomorphism $\rho_V^U : K_U|_V \to K_V$ in $D(\mathcal{O}_V)$,

such that whenever we have $W \subset V \subset U$ with U, V, W in $\mathcal{B}$, then $\rho_W^U = \rho_W^V \circ \rho_V^U|_W$.

We won't be able to prove anything about this without making more assumptions. An interesting case is where $\mathcal{B}$ is a basis for the topology on X. Another is the case where we have a morphism $f : X \to Y$ of topological spaces and the elements of $\mathcal{B}$ are the inverse images of the elements of a basis for the topology of Y.

In Situation 0.1 a *solution* will be a pair (K, ρ_U) where K is an object of $D(\mathcal{O}_X)$ and $\rho_U : K|_U \to K_U$, $U \in \mathcal{B}$ are isomorphisms such that we have $\rho_V^U \circ \rho_U|_V = \rho_V$ for all $V \subset U$, $U, V \in \mathcal{B}$. In certain cases solutions are unique.

Theorem 0.2 (BBD gluing lemma)**.** *In Situation 0.1 assume*

(1) $X = \bigcup_{U \in \mathcal{B}} U$,
(2) *for* $U, V \in \mathcal{B}$ *we have* $U \cap V = \bigcup_{W \in \mathcal{B}, W \subset U \cap V} W$,
(3) *for any* $U \in \mathcal{B}$ *we have* $\mathrm{Ext}^i(K_U, K_U) = 0$ *for* $i < 0$.

Then there exists an object K *of* $D(\mathcal{O}_X)$ *and isomorphisms* $\rho_U : K|_U \to K_U$ *in* $D(\mathcal{O}_U)$ *for* $U \in \mathcal{B}$ *such that* $\rho_V^U \circ \rho_U|_V = \rho_V$ *for all* $V \subset U$ *with* $U, V \in \mathcal{B}$. *The pair* (K, ρ_U) *is unique up to unique isomorphism.*

Lemma 0.3. *Let* $(X, \mathcal{O}_X)$ *be a ringed space. Let* $X = U \cup V$ *be the union of two open subspaces of* X. *Suppose given*

(1) *an object* A *of* $D(\mathcal{O}_U)$,
(2) *an object* B *of* $D(\mathcal{O}_V)$, *and*
(3) *an isomorphism* $c : A|_{U \cap V} \to B|_{U \cap V}$.

Then there exists an object F *of* $D(\mathcal{O}_X)$ *and isomorphisms* $f : F|_U \to A$, $g : F|_V \to B$ *such that* $c = g|_{U \cap V} \circ f^{-1}|_{U \cap V}$. *Moreover, given*

(1) *an object* E *of* $D(\mathcal{O}_X)$,
(2) *a morphism* $a : A \to E|_U$ *of* $D(\mathcal{O}_U)$,
(3) *a morphism* $b : B \to E|_V$ *of* $D(\mathcal{O}_V)$,

such that

$$a|_{U \cap V} = b|_{U \cap V} \circ c.$$

Then there exists a morphism $F \to E$ *in* $D(\mathcal{O}_X)$ *whose restriction to* U *is* $a \circ f$ *and whose restriction to* V *is* $b \circ g$.

Proof. Denote j_U, j_V, $j_{U \cap V}$ the corresponding open immersions. Choose a distinguished triangle

$$F \to Rj_{U,*}A \oplus Rj_{V,*}B \to Rj_{U \cap V,*}(B|_{U \cap V}) \to F[1]$$

where the map $Rj_{V,*}B \to Rj_{U \cap V,*}(B|_{U \cap V})$ is the obvious one and where $Rj_{U,*}A \to Rj_{U \cap V,*}(B|_{U \cap V})$ is the composition of $Rj_{U,*}A \to Rj_{U \cap V,*}(A|_{U \cap V})$ with $Rj_{U \cap V,*}c$. Restricting to U we obtain

$$F|_U \to A \oplus (Rj_{V,*}B)|_U \to (Rj_{U \cap V,*}(B|_{U \cap V}))|_U \to F|_U[1]$$

Denote $j : U \cap V \to U$. Compatibility of restriction to opens and cohomology shows that both $(Rj_{V,*}B)|_U$ and $(Rj_{U \cap V,*}(B|_{U \cap V}))|_U$ are canonically isomorphic

to $Rj_*(B|_{U\cap V})$. Hence the second arrow of the last displayed diagram has a section, and we conclude that the morphism $F|_U \to A$ is an isomorphism. Similarly, the morphism $F|_V \to B$ is an isomorphism. The existence of the morphism $F \to E$ follows from the Mayer-Vietoris sequence for Hom, see Lemma 08BW. □

Lemma 0.4. *Let $f : (X, \mathcal{O}_X) \to (Y, \mathcal{O}_Y)$ be a morphism of ringed spaces. Let $\mathcal{B}$ be a basis for the topology on Y.*

(1) *Assume K is in $D(\mathcal{O}_X)$ such that for $V \in \mathcal{B}$ we have $H^i(f^{-1}(V), K) = 0$ for $i < 0$. Then Rf_*K has vanishing cohomology sheaves in negative degrees, $H^i(f^{-1}(V), K) = 0$ for $i < 0$ for all opens $V \subset Y$, and the rule $V \mapsto H^0(f^{-1}V, K)$ is a sheaf on Y.*
(2) *Assume K, L are in $D(\mathcal{O}_X)$ such that for $V \in \mathcal{B}$ we have $\mathrm{Ext}^i(K|_{f^{-1}V}, L|_{f^{-1}V}) = 0$ for $i < 0$. Then $\mathrm{Ext}^i(K|_{f^{-1}V}, L|_{f^{-1}V}) = 0$ for $i < 0$ for all opens $V \subset Y$ and the rule $V \mapsto \mathrm{Hom}(K|_{f^{-1}V}, L|_{f^{-1}V})$ is a sheaf on Y.*

Proof. Lemma 0D5X tells us $H^i(Rf_*K)$ is the sheaf associated to the presheaf $V \mapsto H^i(f^{-1}(V), K) = H^i(V, Rf_*K)$. The assumptions in (1) imply that Rf_*K has vanishing cohomology sheaves in degrees < 0. We conclude that for any open $V \subset Y$ the cohomology group $H^i(V, Rf_*K)$ is zero for $i < 0$ and is equal to $H^0(V, H^0(Rf_*K))$ for $i = 0$. This proves (1).

To prove (2) apply (1) to the complex $R\mathcal{H}om(K, L)$ using Lemma 08DK to do the translation. □

Lemma 0.5. *In Situation 0.1 assume*

(1) *$X = \bigcup_{U\in\mathcal{B}} U$ and for $U, V \in \mathcal{B}$ we have $U \cap V = \bigcup_{W\in\mathcal{B}, W\subset U\cap V} W$,*
(2) *for any $U \in \mathcal{B}$ we have $\mathrm{Ext}^i(K_U, K_U) = 0$ for $i < 0$.*

If a solution (K, ρ_U) exists, then it is unique up to unique isomorphism and moreover $\mathrm{Ext}^i(K, K) = 0$ for $i < 0$.

Proof. Let (K, ρ_U) and (K', ρ'_U) be a pair of solutions. Let $f : X \to Y$ be the continuous map constructed in Topology, Lemma 0D5Q. Set $\mathcal{O}_Y = f_*\mathcal{O}_X$. Then K, K' and $\mathcal{B}$ are as in Lemma 0.4 part (2). Hence we obtain the vanishing of negative exts for K and we see that the rule

$$V \longmapsto \mathrm{Hom}(K|_{f^{-1}V}, K'|_{f^{-1}V})$$

is a sheaf on Y. As both (K, ρ_U) and (K', ρ'_U) are solutions the maps

$$(\rho'_U)^{-1} \circ \rho_U : K|_U \longrightarrow K'|_U$$

over $U = f^{-1}(f(U))$ agree on overlaps. Hence we get a unique global section of the sheaf above which defines the desired isomorphism $K \to K'$ compatible with all structure available. □

Remark 0.6. With notation and assumptions as in Lemma 0.5. Suppose that $U, V \in \mathcal{B}$. Let $\mathcal{B}'$ be the set of elements of $\mathcal{B}$ contained in $U \cap V$. Then

$$(\{K_{U'}\}_{U'\in\mathcal{B}'}, \{\rho^{U'}_{V'}\}_{V'\subset U' \text{ with } U',V'\in\mathcal{B}'})$$

is a system on the ringed space $U \cap V$ satisfying the assumptions of Lemma 0.5. Moreover, both $(K_U|_{U\cap V}, \rho^U_{U'})$ and $(K_V|_{U\cap V}, \rho^V_{U'})$ are solutions to this system. By the lemma we find a unique isomorphism

$$\rho_{U,V} : K_U|_{U\cap V} \longrightarrow K_V|_{U\cap V}$$

such that for every $U' \subset U \cap V$, $U' \in \mathcal{B}$ the diagram

$$\begin{array}{ccc} K_U|_{U'} & \xrightarrow{\rho_{U,V}|_{U'}} & K_V|_{U'} \\ & {\scriptstyle \rho^U_{U'}} \searrow \quad \swarrow {\scriptstyle \rho^V_{U'}} & \\ & K_{U'} & \end{array}$$

commutes. Pick a third element $W \in \mathcal{B}$. We obtain isomorphisms $\rho_{U,W} : K_U|_{U\cap W} \to K_W|_{U\cap W}$ and $\rho_{V,W} : K_U|_{V\cap W} \to K_W|_{V\cap W}$ satisfying similar properties to those of $\rho_{U,V}$. Finally, we have

$$\rho_{U,W}|_{U\cap V\cap W} = \rho_{V,W}|_{U\cap V\cap W} \circ \rho_{U,V}|_{U\cap V\cap W}$$

This is true by the uniqueness in the lemma because both sides of the equality are the unique isomorphism compatible with the maps $\rho^U_{U''}$ and $\rho^W_{U''}$ for $U'' \subset U\cap V\cap W$, $U'' \in \mathcal{B}$. Some minor details omitted. The collection $(K_U, \rho_{U,V})$ is a descent datum in the derived category for the open covering $\mathcal{U} : X = \bigcup_{U\in\mathcal{B}} U$ of X. In this language we are looking for "effectiveness of the descent datum" when we look for the existence of a solution.

Lemma 0.7. *In Situation 0.1 assume*

(1) $X = U_1 \cup \ldots \cup U_n$ *with* $U_i \in \mathcal{B}$,
(2) *for* $U, V \in \mathcal{B}$ *we have* $U \cap V = \bigcup_{W\in\mathcal{B}, W\subset U\cap V} W$,
(3) *for any* $U \in \mathcal{B}$ *we have* $\mathrm{Ext}^i(K_U, K_U) = 0$ *for* $i < 0$.

Then a solution exists and is unique up to unique isomorphism.

Proof. Uniqueness was seen in Lemma 0.5. We may prove the lemma by induction on n. The case $n = 1$ is immediate.

The case $n = 2$. Consider the isomorphism $\rho_{U_1,U_2} : K_{U_1}|_{U_1\cap U_2} \to K_{U_2}|_{U_1\cap U_2}$ constructed in Remark 0.6. By Lemma 0.3 we obtain an object K in $D(\mathcal{O}_X)$ and isomorphisms $\rho_{U_1} : K|_{U_1} \to K_{U_1}$ and $\rho_{U_2} : K|_{U_2} \to K_{U_2}$ compatible with ρ_{U_1,U_2}. Take $U \in \mathcal{B}$. We will construct an isomorphism $\rho_U : K|_U \to K_U$ and we will leave it to the reader to verify that (K, ρ_U) is a solution. Consider the set $\mathcal{B}'$ of elements of $\mathcal{B}$ contained in either $U\cap U_1$ or contained in $U\cap U_2$. Then $(K_U, \rho^U_{U'})$ is a solution for the system $(\{K_{U'}\}_{U'\in\mathcal{B}'}, \{\rho^{U'}_{V'}\}_{V'\subset U' \text{ with } U',V'\in\mathcal{B}'})$ on the ringed space U. We claim that $(K|_U, \tau_{U'})$ is another solution where $\tau_{U'}$ for $U' \in \mathcal{B}'$ is chosen as follows: if $U' \subset U_1$ then we take the composition

$$K|_{U'} \xrightarrow{\rho_{U_1}|_{U'}} K_{U_1}|_{U'} \xrightarrow{\rho^{U_1}_{U'}} K_{U'}$$

and if $U' \subset U_2$ then we take the composition

$$K|_{U'} \xrightarrow{\rho_{U_2}|_{U'}} K_{U_2}|_{U'} \xrightarrow{\rho^{U_2}_{U'}} K_{U'}.$$

To verify this is a solution use the property of the map ρ_{U_1,U_2} described in Remark 0.6 and the compatibility of ρ_{U_1} and ρ_{U_2} with ρ_{U_1,U_2}. Having said this we apply Lemma 0.5 to see that we obtain a unique isomorphism $K|_{U'} \to K_{U'}$ compatible with the maps $\tau_{U'}$ and $\rho^U_{U'}$ for $U' \in \mathcal{B}'$.

The case $n > 2$. Consider the open subspace $X' = U_1 \cup \ldots \cup U_{n-1}$ and let $\mathcal{B}'$ be the set of elements of $\mathcal{B}$ contained in X'. Then we find a system $(\{K_U\}_{U\in\mathcal{B}'}, \{\rho^U_V\}_{U,V\in\mathcal{B}'})$ on the ringed space X' to which we may apply our induction hypothesis. We find a solution $(K_{X'}, \rho^{X'}_U)$. Then we can consider the collection $\mathcal{B}^* = \mathcal{B} \cup \{X'\}$ of opens of X and we see that we obtain a system $(\{K_U\}_{U\in\mathcal{B}^*}, \{\rho^U_V\}_{V\subset U \text{ with } U,V\in\mathcal{B}^*})$. Note that this new system also satisfies condition (3) by Lemma 0.5 applied to the solution $K_{X'}$. For this system we have $X = X' \cup U_n$. This reduces us to the case $n = 2$ we worked out above. □

Lemma 0.8. *Let X be a ringed space. Let E be a well ordered set and let*

$$X = \bigcup\nolimits_{\alpha\in E} W_\alpha$$

be an open covering with $W_\alpha \subset W_{\alpha+1}$ and $W_\alpha = \bigcup_{\beta<\alpha} W_\beta$ if α is not a successor. Let K_α be an object of $D(\mathcal{O}_{W_\alpha})$ with $\mathrm{Ext}^i(K_\alpha, K_\alpha) = 0$ for $i < 0$. Assume given isomorphisms $\rho^\alpha_\beta : K_\alpha|_{W_\beta} \to K_\beta$ in $D(\mathcal{O}_{W_\beta})$ for all $\beta < \alpha$ with $\rho^\alpha_\gamma = \rho^\beta_\gamma \circ \rho^\alpha_\beta|_{W_\gamma}$ for $\gamma < \beta < \alpha$. Then there exists an object K in $D(\mathcal{O}_X)$ and isomorphisms $K|_{W_\alpha} \to K_\alpha$ for $\alpha \in E$ compatible with the isomorphisms ρ^α_β.

Proof. In this proof $\alpha, \beta, \gamma, \ldots$ represent elements of E. Choose a K-injective complex $I^\bullet_\alpha$ on W_α representing K_α. For $\beta < \alpha$ denote $j_{\beta,\alpha} : W_\beta \to W_\alpha$ the inclusion morphism. Using transfinite recursion we will construct for all $\beta < \alpha$ a map of complexes

$$\tau_{\beta,\alpha} : (j_{\beta,\alpha})_! I^\bullet_\beta \longrightarrow I^\bullet_\alpha$$

representing the adjoint to the inverse of the isomorphism $\rho^\alpha_\beta : K_\alpha|_{W_\beta} \to K_\beta$. Moreover, we will do this in such that for $\gamma < \beta < \alpha$ we have

$$\tau_{\gamma,\alpha} = \tau_{\beta,\alpha} \circ (j_{\beta,\alpha})_! \tau_{\gamma,\beta}$$

as maps of complexes. Namely, suppose already given $\tau_{\gamma,\beta}$ composing correctly for all $\gamma < \beta < \alpha$. If $\alpha = \alpha' + 1$ is a successor, then we choose any map of complexes

$$(j_{\alpha',\alpha})_! I^\bullet_{\alpha'} \to I^\bullet_\alpha$$

which is adjoint to the inverse of the isomorphism $\rho^\alpha_{\alpha'} : K_\alpha|_{W_{\alpha'}} \to K_{\alpha'}$ (possible because $I^\bullet_\alpha$ is K-injective) and for any $\beta < \alpha'$ we set

$$\tau_{\beta,\alpha} = \tau_{\alpha',\alpha} \circ (j_{\alpha',\alpha})_! \tau_{\beta,\alpha'}$$

If α is not a successor, then we can consider the complex on W_α given by

$$C^\bullet = \mathrm{colim}_{\beta<\alpha} (j_{\beta,\alpha})_! I^\bullet_\beta$$

(termwise colimit) where the transition maps of the sequence are given by the maps $\tau_{\beta',\beta}$ for $\beta' < \beta < \alpha$. We claim that $C^\bullet$ represents K_α. Namely, for $\beta < \alpha$ the restriction of the coprojection $(j_{\beta,\alpha})_! I^\bullet_\beta \to C^\bullet$ gives a map

$$\sigma_\beta : I^\bullet_\beta \longrightarrow C^\bullet|_{W_\beta}$$

which is a quasi-isomorphism: if $x \in W_\beta$ then looking at stalks we get

$$(C^\bullet)_x = \mathrm{colim}_{\beta'<\alpha} \left((j_{\beta',\alpha})_! I^\bullet_{\beta'}\right)_x = \mathrm{colim}_{\beta\le\beta'<\alpha} (I^\bullet_{\beta'})_x \longleftarrow (I^\bullet_\beta)_x$$

which is a quasi-isomorphism. Here we used that taking stalks commutes with colimits, that filtered colimits are exact, and that the maps $(I^\bullet_\beta)_x \to (I^\bullet_{\beta'})_x$ are quasi-isomorphisms for $\beta \le \beta' < \alpha$. Hence $(C^\bullet, \sigma_\beta^{-1})$ is a solution to the system $(\{K_\beta\}_{\beta<\alpha}, \{\rho^\beta_{\beta'}\}_{\beta'<\beta<\alpha})$. Since $(K_\alpha, \rho^\alpha_\beta)$ is another solution we obtain a unique isomorphism $\sigma : K_\alpha \to C^\bullet$ in $D(\mathcal{O}_{W_\alpha})$ compatible with all our maps, see Lemma 0.7 (this is where we use the vanishing of negative ext groups). Choose a morphism $\tau : C^\bullet \to I^\bullet_\alpha$ of complexes representing σ. Then we set

$$\tau_{\beta,\alpha} = \tau|_{W_\beta} \circ \sigma_\beta$$

to get the desired maps. Finally, we take K to be the object of the derived category represented by the complex

$$K^\bullet = \mathrm{colim}_{\alpha\in E} (W_\alpha \to X)_! I^\bullet_\alpha$$

where the transition maps are given by our carefully constructed maps $\tau_{\beta,\alpha}$ for $\beta < \alpha$. Arguing exactly as above we see that for all α the restriction of the coprojection determines an isomorphism

$$K|_{W_\alpha} \longrightarrow K_\alpha$$

compatible with the given maps ρ^α_β. □

Proof. A pair (K, ρ_U) is called a solution in the text above. The uniqueness follows from Lemma 0.5. If X has a finite covering by elements of $\mathcal{B}$ (for example if X is quasi-compact), then the theorem is a consequence of Lemma 0.7. In the general case we argue in exactly the same manner, using transfinite induction and Lemma 0.8.

First we use transfinite recursion to choose opens $W_\alpha \subset X$ for any ordinal α. Namely, we set $W_0 = \emptyset$. If $\alpha = \beta + 1$ is a successor, then either $W_\beta = X$ and we set $W_\alpha = X$ or $W_\beta \neq X$ and we set $W_\alpha = W_\beta \cup U_\alpha$ where $U_\alpha \in \mathcal{B}$ is not contained in W_β. If α is a limit ordinal we set $W_\alpha = \bigcup_{\beta<\alpha} W_\beta$. Then for large enough α we have $W_\alpha = X$. Observe that for every α the open W_α is a union of elements of $\mathcal{B}$. Hence if $\mathcal{B}_\alpha = \{U \in \mathcal{B}, U \subset W_\alpha\}$, then

$$S_\alpha = (\{K_U\}_{U\in\mathcal{B}_\alpha}, \{\rho^U_V\}_{V\subset U \text{ with } U,V\in\mathcal{B}_\alpha})$$

is a system as in Lemma 0.5 on the ringed space W_α.

We will show by transfinite induction that for every α the system S_α has a solution. This will prove the theorem as this system is the system given in the theorem for large α.

The case where $\alpha = \beta + 1$ is a successor ordinal. (This case was already treated in the proof of the lemma above but for clarity we repeat the argument.) Recall that $W_\alpha = W_\beta \cup U_\alpha$ for some $U_\alpha \in \mathcal{B}$ in this case. By induction hypothesis we have a solution $(K_{W_\beta}, \{\rho_U^{W_\beta}\}_{U\in\mathcal{B}_\beta})$ for the system S_β. Then we can consider the collection $\mathcal{B}^*_\alpha = \mathcal{B}_\alpha \cup \{W_\beta\}$ of opens of W_α and we see that we obtain a system $(\{K_U\}_{U\in\mathcal{B}^*_\alpha}, \{\rho^U_V\}_{V\subset U \text{ with } U,V\in\mathcal{B}^*_\alpha})$. Note that this new system also satisfies condition (3) by Lemma 0.5 applied to the solution K_{W_β}. For this system we have $W_\alpha = W_\beta \cup U_\alpha$. This reduces us to the case handled in Lemma 0.7.

The case where α is a limit ordinal. Recall that $W_\alpha = \bigcup_{\beta<\alpha} W_\beta$ in this case. For $\beta < \alpha$ let $(K_{W_\beta}, \{\rho_U^{W_\beta}\}_{U\in\mathcal{B}_\beta})$ be the solution for S_β. For $\gamma < \beta < \alpha$ the restriction $K_{W_\beta}|_{W_\gamma}$ endowed with the maps $\rho_U^{W_\beta}$, $U \in \mathcal{B}_\gamma$ is a solution for S_γ. By uniqueness we get unique isomorphisms $\rho^{W_\beta}_{W_\gamma} : K_{W_\beta}|_{W_\gamma} \to K_{W_\gamma}$ compatible with the maps $\rho_U^{W_\beta}$ and $\rho_U^{W_\gamma}$ for $U \in \mathcal{B}_\gamma$. These maps compose in the correct manner, i.e., $\rho^{W_\gamma}_{W_\delta} \circ \rho^{W_\beta}_{W_\gamma}|_{W_\delta} = \rho^{W_\delta}_{W_\beta}$ for $\delta < \gamma < \beta < \alpha$. Thus we may apply Lemma 0.8 (note that the vanishing of negative exts is true for K_{W_β} by Lemma 0.5 applied to the solution K_{W_β}) to obtain K_{W_α} and isomorphisms

$$\rho^{W_\alpha}_{W_\beta} : K_{W_\alpha}|_{W_\beta} \longrightarrow K_{W_\beta}$$

compatible with the maps $\rho^{W_\beta}_{W_\gamma}$ for $\gamma < \beta < \alpha$.

To show that K_{W_α} is a solution we still need to construct the isomorphisms $\rho_U^{W_\alpha} : K_{W_\alpha}|_U \to K_U$ for $U \in \mathcal{B}_\alpha$ satisfying certain compatibilities. We choose $\rho_U^{W_\alpha}$ to be the unique map such that for any $\beta < \alpha$ and any $V \in \mathcal{B}_\beta$ with $V \subset U$ the diagram

$$\begin{array}{ccc} K_{W_\alpha}|_V & \xrightarrow{\rho_U^{W_\alpha}|_V} & K_U|_V \\ {\scriptstyle \rho^{W_\alpha}_{W_\beta}|_V}\downarrow & & \downarrow{\scriptstyle \rho^V_U} \\ K_{W_\beta} & \xrightarrow{\rho_V^{W_\beta}} & K_V \end{array}$$

commutes. This makes sense because

$$(\{K_V\}_{V\subset U, V\in\mathcal{B}_\beta \text{ for some } \beta<\alpha}, \{\rho^{V'}_V\}_{V\subset V' \text{ with } V,V'\subset U \text{ and } V,V'\in\mathcal{B}_\beta \text{ for some } \beta<\alpha})$$

is a system as in Lemma 0.5 on the ringed space U and because (K_U, ρ_V^U) and $(K_{W_\alpha}|_U, \rho_V^{W_\beta} \circ \rho_{W_\beta}^{W_\alpha}|_V)$ are both solutions for this system. This gives existence and uniqueness. We omit the proof that these maps satisfy the desired compatibilities (it is just bookkeeping). □